



LAN Scanner Pro

(Enhanced Network Discovery & Security Tool)

A PROJECT REPORT

for the subject

PRJT 411

Submitted By

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ANAND-388110

MONTH, YEAR

COLLEGE OF AGRICULTURAL INFORMATION TECHNOLOGY
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DECLARATION BY THE STUDENT

I declare that the project entitled: **Lan Scanner Pro** is submitted by **Maheshwari Vraj L. (3060822017)** under the supervision of: **Dr.Y.R.Ghodasara** for award of Bachelor of Agricultural Information Technology Degree of Anand Agricultural University carried out during semester **7th** for the subject of **PRJT 411** embodies my original work.

Name of the Student: **Maheshwari Vraj L.**

Signature: _____

CERTIFICATE

This is to certify that the project work entitled **Lan Scanner Pro** for the subject **PRJT 411** has been carried out by **Maheshwari Vraj L** under my guidance in fulfillment of the degree of Bachelor of Agricultural Information Technology of College of Agricultural Information Technology (7th Semester) of Anand Agricultural University, Anand during the academic year 2025-26.

Date: _____

Place: **College Of Agricultural Information Technology, AAU, Anand**

Name of the Guide: **Dr.Y.R.Ghodasara**

Signature: _____

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1. Project Title

LAN Scanner Pro – Enhanced Network Discovery & Security Tool

2. Introduction

Today, monitoring and analysis of the devices connected over a LAN are indispensable parts of security, performance, and maintenance in modern computer networks.

Traditional command-line tools such as ping, nmap, and netstat require technical expertise and are not user-friendly for end-users.

LAN Scanner Pro is an advanced, Python-based desktop application that provides a user-friendly interface to simplify LAN scanning. Its feature list includes support for multi-threaded scans, port detection, device fingerprinting, topology visualization, and vulnerability hints, all with reports that can be exported using Tkinter.

The system is ideal for:

- Network administrators
- Cybersecurity students
- IT professionals
- Enthusiasts learning network scanning & security

3. Purpose of the Project

The primary purpose of LAN Scanner Pro is to provide a fast, efficient, and user-friendly network scanning tool that eliminates the complexity of command-line utilities.

It aims to:

- Automate device discovery within a LAN
- Provide more information than basic scanning tools
- Offer security insights based on detected services
- Support visual analysis using topology diagrams and graphs
- Enable administrators to export detailed scan reports

4. Objectives

To simplify LAN scanning for beginners and professionals.

To detect active hosts and open ports in real time.

To identify running services using banner grabbing and DNS lookup.

To provide basic vulnerability hints for security improvement.

To generate visual reports, topology maps, and statistical charts.

To offer fast multi-threaded scanning for improved performance.

5. Modules of the System

1. Scan Configuration Module

- Set IP range (single IP, CIDR, or custom range)
- Choose common ports, specific ports, or custom port ranges
- Configure timeout, workers, banner grabbing, reverse DNS, etc.

2. Network Scanning Module

- Multi-threaded active host detection
- Open port scanning

3. Reverse DNS & Banner Grabbing Module

- Hostname identification

4. Vulnerability Assessment Module

- Security hints based on open ports

5. Results Visualization Module

- Network topology map
- Statistical charts
- Tabbed interface for organized results

6. Reporting & Export Module

- Export results in CSV, JSON, HTML, TXT
- Save/load configuration profiles

7. Logging Module

- Timestamped action logs
- Error logs
- Scan progress tracking

6. Platform, Hardware & Software Used

Hardware Requirements

- Processor: Dual Core or higher
- RAM: Minimum 4 GB
- Storage: 100 MB free space
- Network Adapter: LAN/Wi-Fi capable

Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.7 or above
- GUI Framework: Tkinter
- Libraries / Packages:
 - socket – network communication
 - ipaddress – IP range management
 - threading – multi-threaded scanning
 - matplotlib – graphs and charts
 - ping3 – ping testing
 - csv, json – data export

Development Platform

- Python IDLE / VS Code / PyCharm

7. Functionality of the System

1. User Input & Configuration

- User selects IP range and port options.
- Additional options like banner grabbing, DNS lookup, and vulnerability checks.

2. Multi-Threaded Scanning

- Application scans multiple hosts and ports simultaneously.
- Detects active/inactive hosts quickly.

3. Service & Port Analysis

- Detects open ports on each host.
- Banners retrieved from services like:
 - HTTP/HTTPS
 - FTP
 - SSH
 - SMTP, etc.

4. Network Discovery & Fingerprinting

- Identifies hostnames.

5. Topology Mapping

- Visualizes discovered hosts in a network graph.

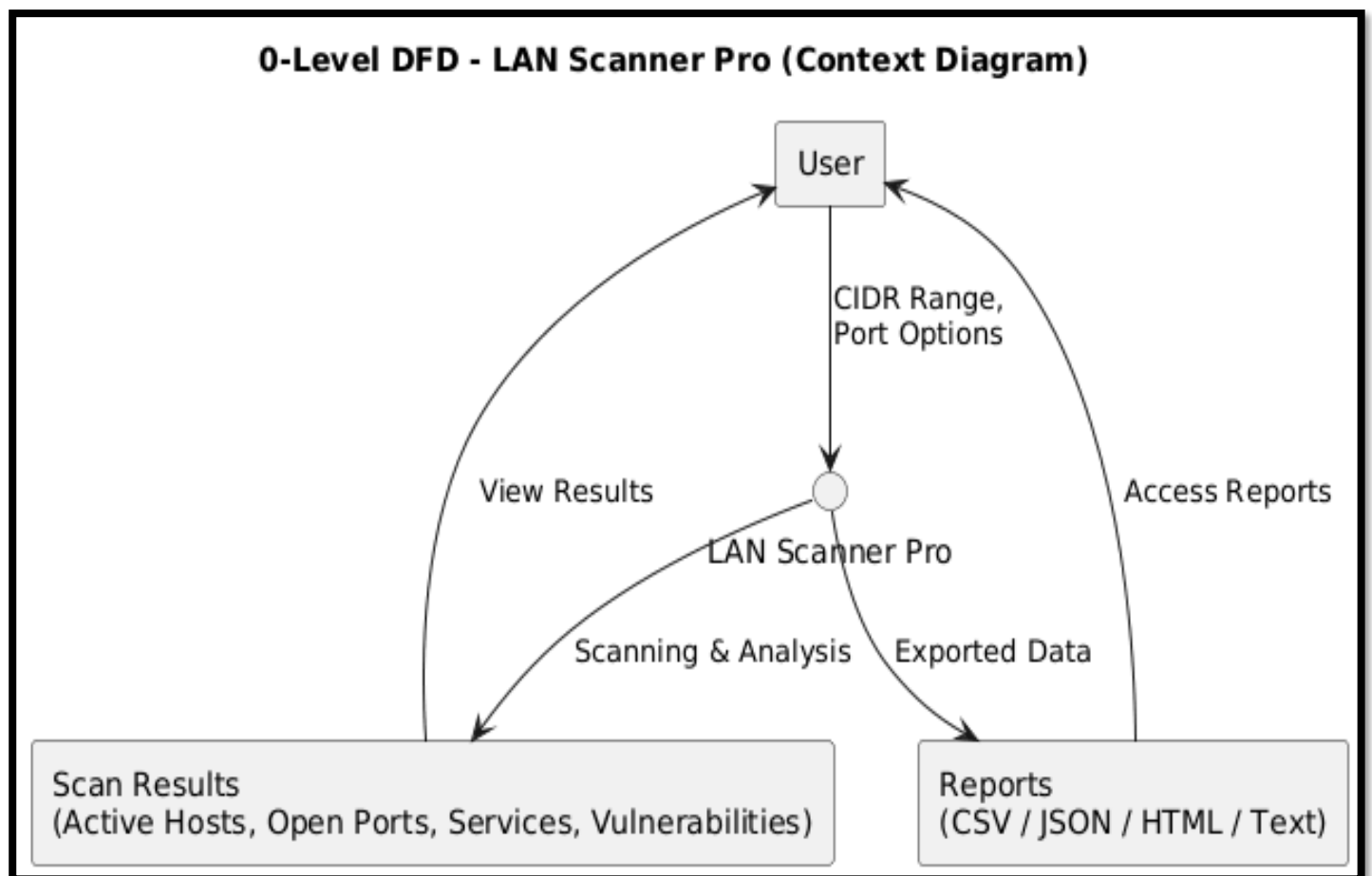
6. Security Assessment

- Provides security hints for:

7. Logging & Reporting

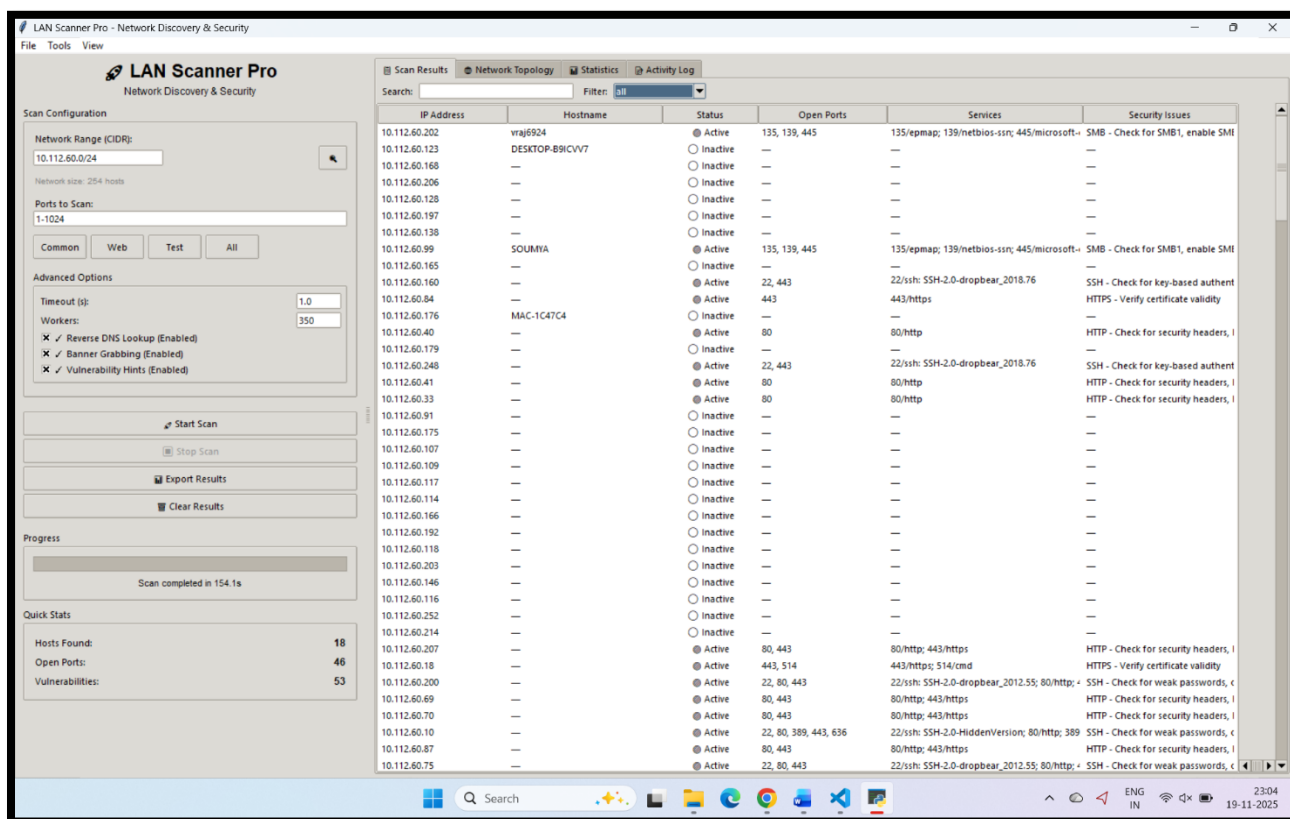
- Activity logs with timestamps
- Exports reports in multiple formats
- Supports profile saving for repeat scans

8. Data Flow Diagram

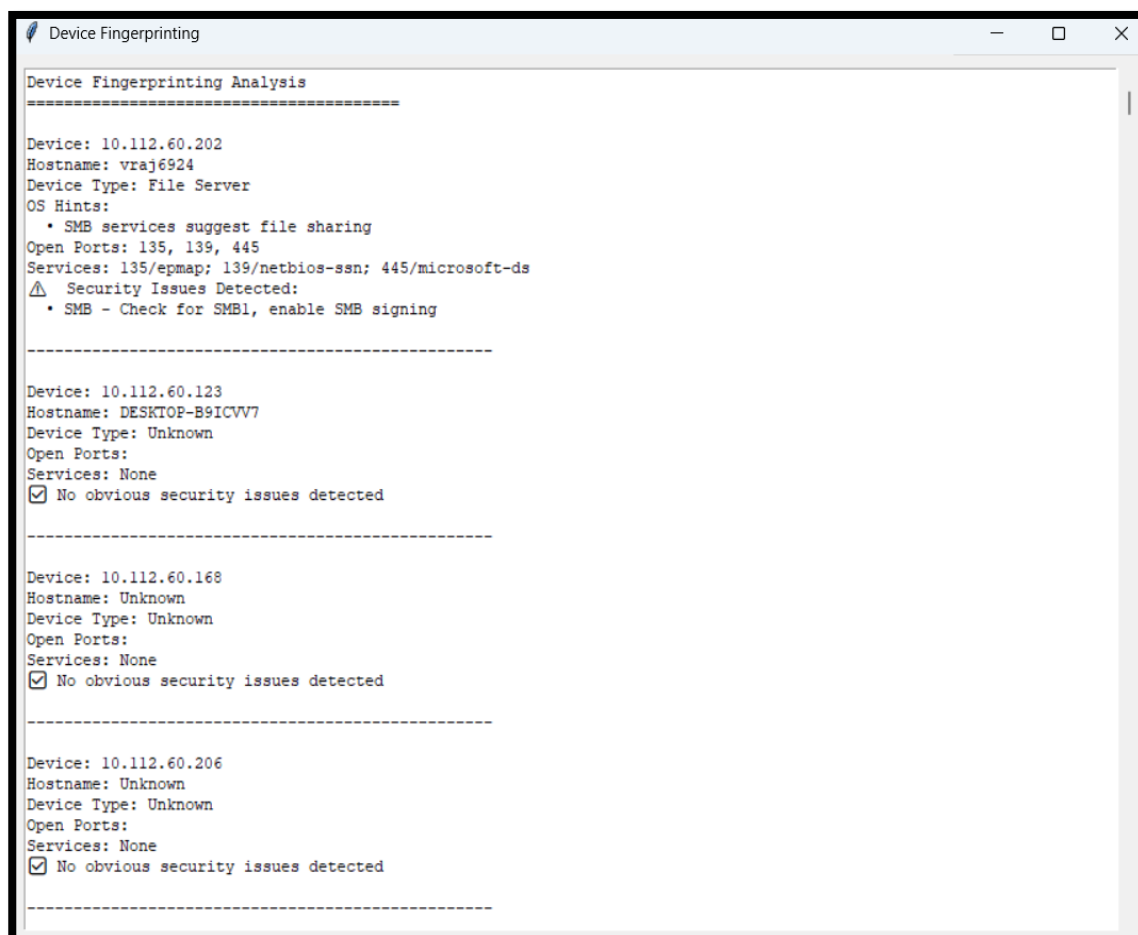


9. screen shots

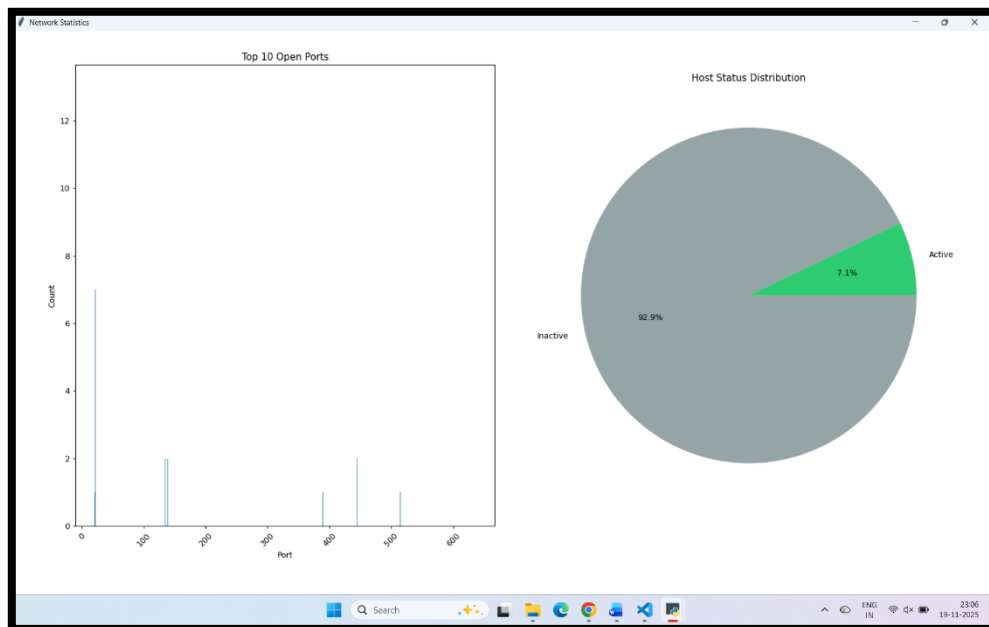
- Main Screen/Home Screen



- Device Fingerprint



- **Network Statics**



- **Network topology**

